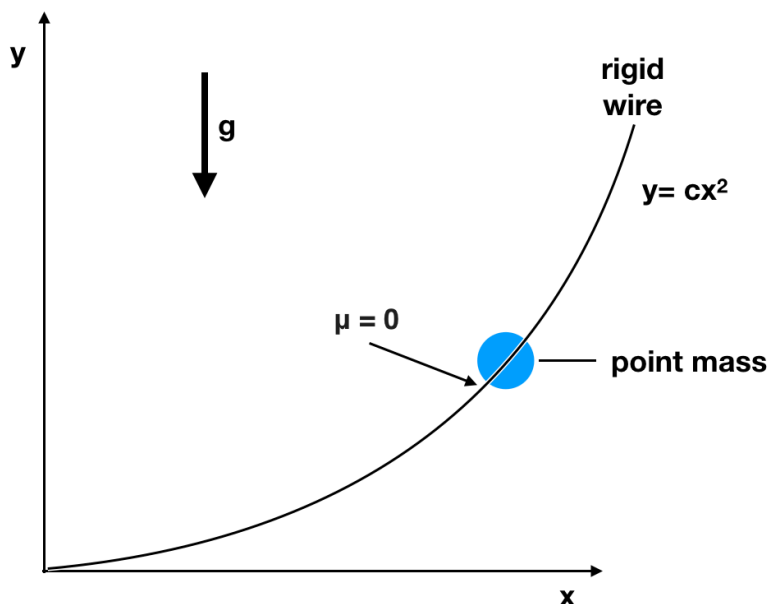


## Problem 28 - Bead on a wire using Lagrange Equations

By Daniel Sachs

(a) Wire equation  $y = cx^2$ :

**Q: Find  $\ddot{x}$   
using  
Lagrange  
Equations**



Find  $\dot{y}$

$$y = cx^2; \dot{y} = c2\dot{x}x$$

Find  $E_k$  &  $E_p$

$$E_k = \frac{1}{2}mv^2 = \frac{1}{2}m(\dot{y}^2 + \dot{x}^2) = \frac{1}{2}m([c2\dot{x}x]^2 + \dot{x}^2) = \frac{1}{2}m\dot{x}^2(4c^2x^2 + 1)$$

$$E_p = mgy = mgcx^2$$

Plug into the Lagrange Equation and solve for  $\ddot{x}$

$$\mathcal{L} = E_k - E_p = \frac{1}{2}m\dot{x}^2(4c^2x^2 + 1) - mgcx^2$$

$$\frac{d}{dt}\left(\frac{d\mathcal{L}}{d\dot{x}}\right) - \frac{d\mathcal{L}}{dx} = 0$$

$$\frac{d\mathcal{L}}{d\dot{x}} = m\dot{x}(4c^2x^2 + 1)$$

$$\frac{d}{dt}\left(\frac{d\mathcal{L}}{d\dot{x}}\right) = m\ddot{x}(4c^2x^2 + 1) + m8c^2x\dot{x}^2$$

$$\frac{d\mathcal{L}}{dx} = 4mc^2\dot{x}^2x - 2mgcx$$

$$\frac{d}{dt}\left(\frac{d\mathcal{L}}{d\dot{x}}\right) - \frac{d\mathcal{L}}{dx} = m\ddot{x}(4c^2x^2 + 1) + m8c^2x\dot{x}^2 - 4mc^2\dot{x}^2x + 2mgcx = 0$$

$$\ddot{x}(4c^2x^2 + 1) = -8c^2x\dot{x}^2 + 4c^2\dot{x}^2x - 2gcx$$

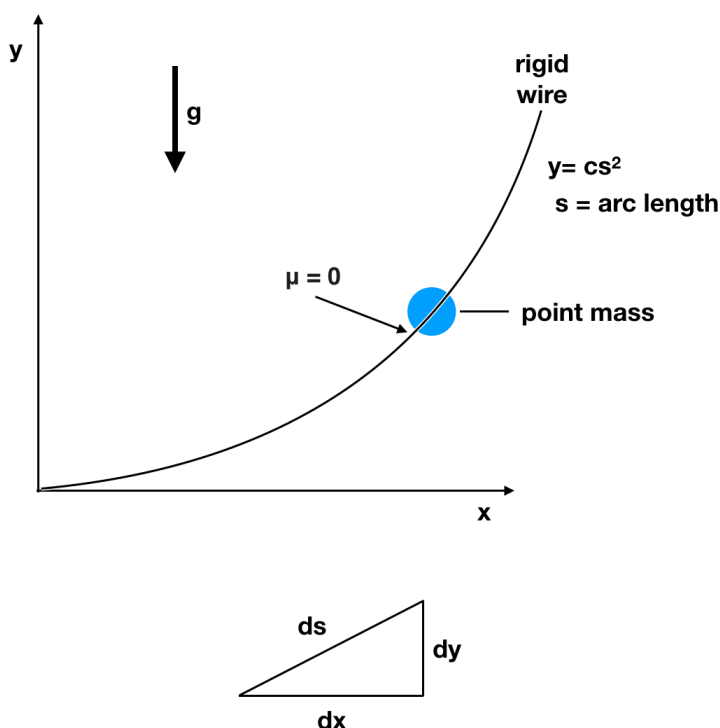
## Problem 28 - Bead on a wire using Lagrange Equations

$$\ddot{x} = \frac{-4c^2 x \dot{x}^2 - 2g c x}{4c^2 x^2 + 1}$$

We can compare this equation to that found in the solution problem 26 page 3. The two equations are identical.

(b) Wire equation  $y = cs^2$ :

**Q: Find  $\ddot{s}$  using Lagrange Equations**



Find  $v$

$y = cs^2$ ;  $v = \dot{s}$  Since  $S$  is the arc length

Find  $E_k$  &  $E_p$

$$E_k = \frac{1}{2}mv^2 = \frac{1}{2}m\dot{s}^2$$

$$E_p = mgy = mgcs^2$$

Plug into the Lagrange Equation and solve for  $\ddot{s}$

$$\mathcal{L} = E_k - E_p = \frac{1}{2}m\dot{s}^2 - mgcs^2$$

## Problem 28 - Bead on a wire using Lagrange Equations

$$\frac{d}{dt} \left( \frac{d\mathcal{L}}{d\dot{s}} \right) - \frac{d\mathcal{L}}{dx} = 0$$

$$\frac{d\mathcal{L}}{d\dot{s}} = m\dot{s}$$

$$\frac{d}{dt} \left( \frac{d\mathcal{L}}{d\dot{s}} \right) = m\ddot{s}$$

$$\frac{d\mathcal{L}}{ds} = -2mgcs$$

$$\frac{d}{dt} \left( \frac{d\mathcal{L}}{d\dot{s}} \right) - \frac{d\mathcal{L}}{dx} = m\ddot{s} + 2mgcs = 0$$

$$\ddot{s} = -2gcs$$

Extra credits

i) Draw the curve  $y = cs^2$ ,  $\dot{s} = \sqrt{\dot{x}^2 + \dot{y}^2}$ :

Given  $s$  and  $\dot{s}$  we can solve for  $\dot{y}$  and  $\dot{x}$  inside a numerically. From  $\dot{s}, \dot{y}, \dot{x}$  we will find  $s, y, x$ .

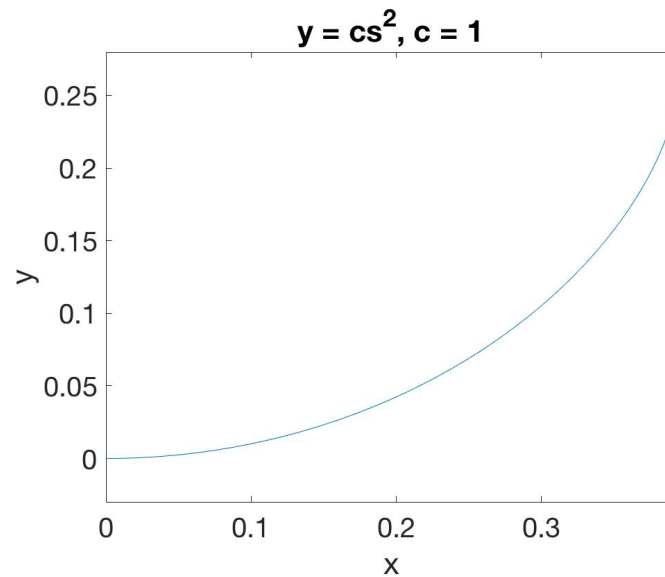
$$\dot{y} = c2\dot{s}s; \dot{x} = \sqrt{\dot{s}^2 - \dot{y}^2}$$

We can manipulate the equations above into the form for an RHS, integrating with initial conditions of  $x = 0$  and  $y = 0$  we get the plot below. Code is in the appendix.

$$\frac{\dot{y}}{\dot{s}} = \frac{dy}{ds} = c2s$$

$$\frac{dx}{ds} = \frac{\dot{x}}{\dot{s}} = \frac{\sqrt{\dot{s}^2 - \dot{y}^2}}{\dot{s}} = \sqrt{1 - \left( \frac{dy}{ds} \right)^2}$$

## Problem 28 - Bead on a wire using Lagrange Equations



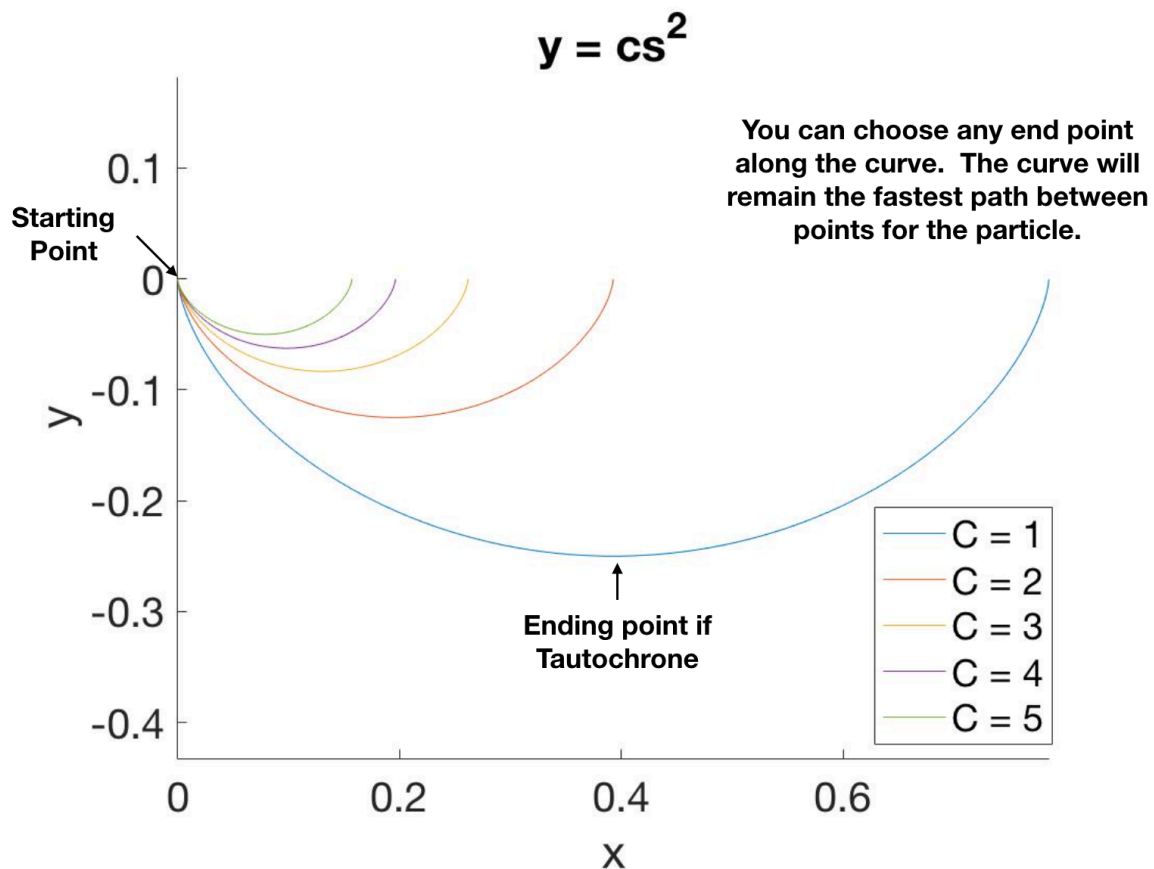
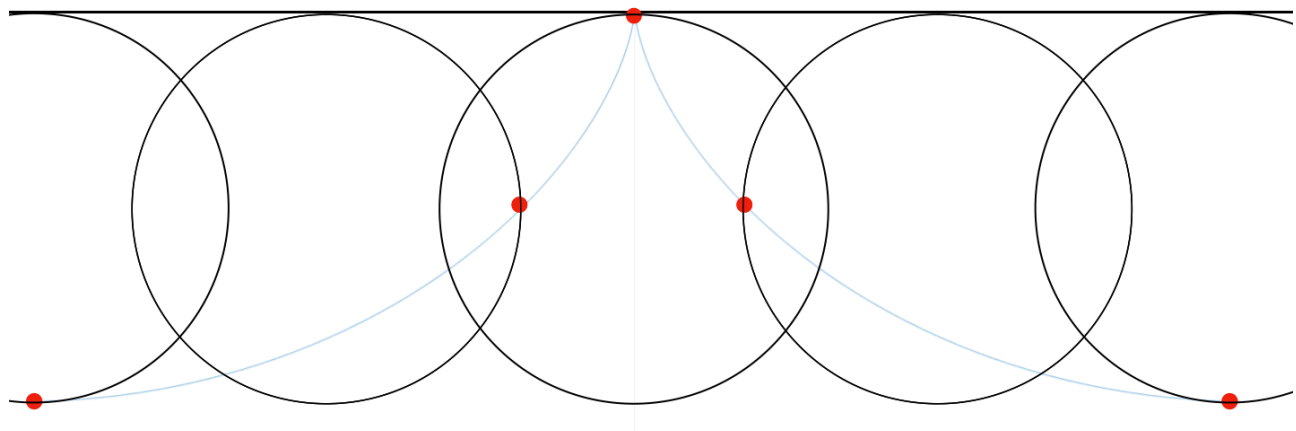
ii) What is the curve?

This curve has three names Cycloid which has two useful subgroups of Brachistochrone, and Tautochrone.

**Brachistochrone:** The fastest decent of the bead when the starting point is not directly above the end point. The curve can take one full rotation of the circle such that the start and end points are at the same level

**Tautochrone:** The curve so that the time to reach the bottom is indemnity of starting position assuming no friction and uniform gravity. The curve ends with a horizontal tangent. This is the curve looked at in the problem.

## Problem 28 - Bead on a wire using Lagrange Equations



Drawing a cycloid Tautochrone between two Cartesian points. Since  $y = cs^2$  will require using a numerical solver to get between the two points so we will use an alternative form of the equations.

$$X_2 - X_1 = r(t + \sin(t)); \quad Y_2 - Y_1 = r(1 - \cos(t));$$

## Problem 28 - Bead on a wire using Lagrange Equations

We can solve for the  $r$  and  $t$  with the system of equations above. Then plug it into the equations below.

$$X = r(t + \sin(T)) + X_1; \quad Y_1 = r(1 - \cos(T)) + Y_1; \quad 0 \leq T \leq t$$

### Appendix

Code for plotting  $y = cs^2$

```
%MAE 4730
%HW Problem 28 part 2
close all; clear all; clc

c = 1; %c input

ic = [0 0]; %s(0)=0; y(0)=0; x(0)=0;
s = [0 .5]; %time span

options = odeset('RelTol',1e-9,'AbsTol',1e-9); %tolerance
[~,output]=ode45(@RHS,s,ic,options,c);

%formats outputs
x = output(:,1);
y = output(:,2);

%makes plot
plot(x,y)
set(0,'DefaultAxesFontSize', 20)
axis('equal')
title('y = cs^2, c = 1')
xlabel('x')
ylabel('y')

function [dn] = RHS(s,y,c)
dy = c*2*s; %finds dy
dx = sqrt(1-dy^2); %finds dx
dn = [dx; dy];
end
```